

CS4800: Designing Sustainable ICT Systems

Lecture 6: Sustainable Electronics Design

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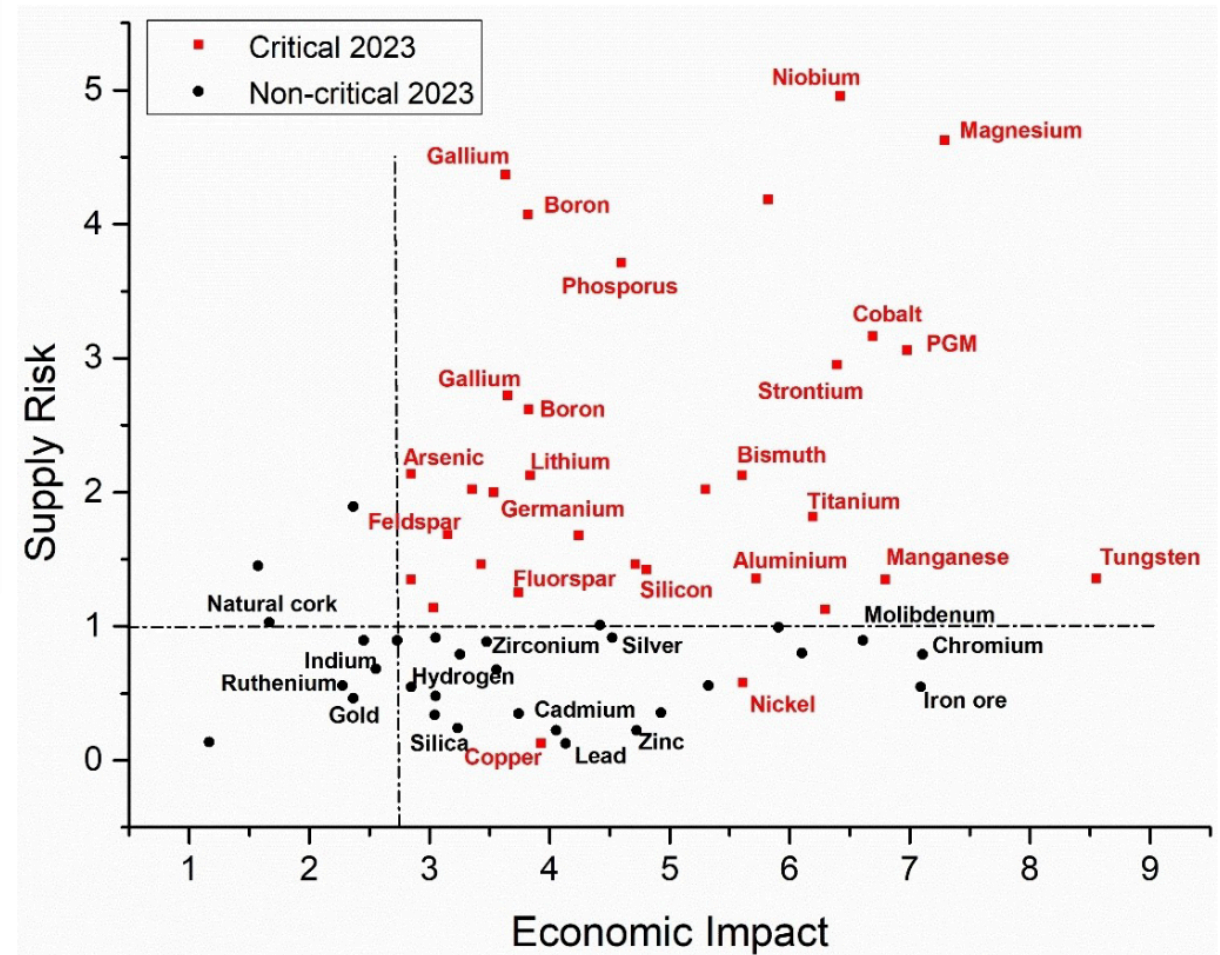
What will you learn in this lecture?

Lecture 6: Sustainable Electronics Design

- Material basis of ICT systems
- Critical raw materials, recycling
- Principles of sustainable electronics design
- PCB design principles
- Battery-free electronics
- Digital product passports

Reminder: critical raw materials

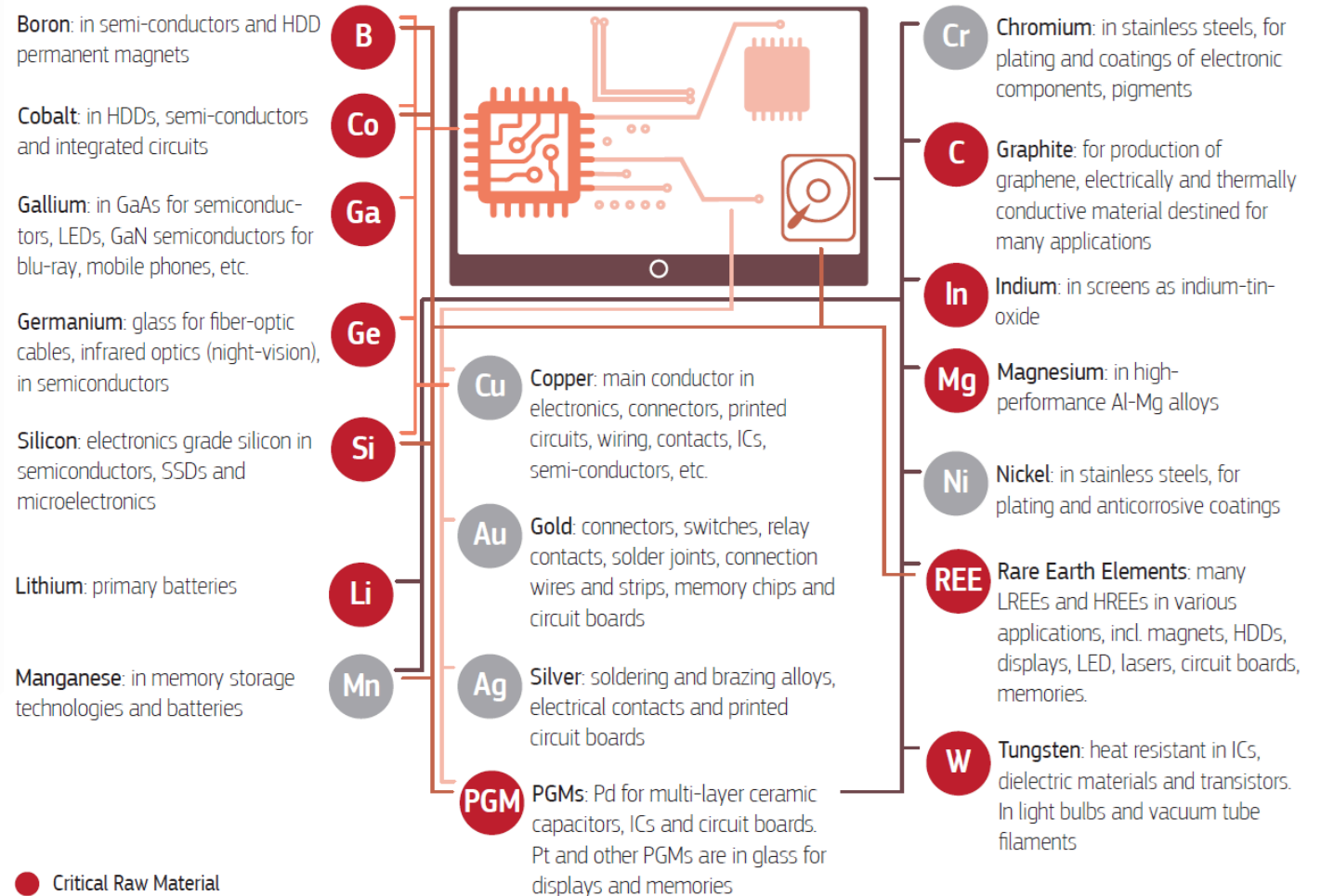
- Raw materials form basis of manufacturing economy
- [Criticality](#): high economic importance, high risk of supply chain disruption
 - Economic importance: how much of economy relies on it?
 - Supply risk: how many potential suppliers are there?
- [General policy concern](#) for EU (energy transition, globalization), but especially important for ICT because...



[European Commission 2023 list of critical raw materials](#)

Material basis of ICT

- Modern electronics contain dozens of elements
- Metals: finite, increasingly inaccessible deposits; environmental/social impact of mining
- Plastics: 99% of world demand covered by petrochemicals



Critical raw materials (red) in modern electronics

Material basis of ICT

What's in your laptop?

1. [Circuit board](#): fiberglass (epoxy, polyester resin, glass fiber) or polyimide monomer substrate, copper plate layer, gold, silver, or tin for soldering
2. [Hard disk drive](#): neodymium, iron, and boron in magnets, cobalt, chromium, and platinum in data layers, nickel and tantalum in buffer layers, rubidium layers, chromium, titanium, and tantalum in seed layers, polymer lubricant and glass substrate
3. [Flash memory](#): tantalum, aluminum oxide, and silicon oxide
4. [CPU/GPU](#): silicon, tantalum and palladium transistors, copper, boron, tungsten, hafnium
5. [Screen](#): indium, glass
6. [Battery](#): copper collector foil, graphite anode, titanium and niobium catalysts, cobalt or lithium cathodes, manganese and nickel

Recycling: potential

Why not just recycle everything at end of life?

How much of each material in electronics is [currently](#) recycled?

- Iron (used both in small-scale electronics and as a structural element in steel)
- Aluminum (same as above)
- Noble metals, silver, gold, and palladium
- Scarce metals: germanium, gallium, rare earth metals, tantalum
- Lithium, cobalt
- Plastics

Recycling: potential

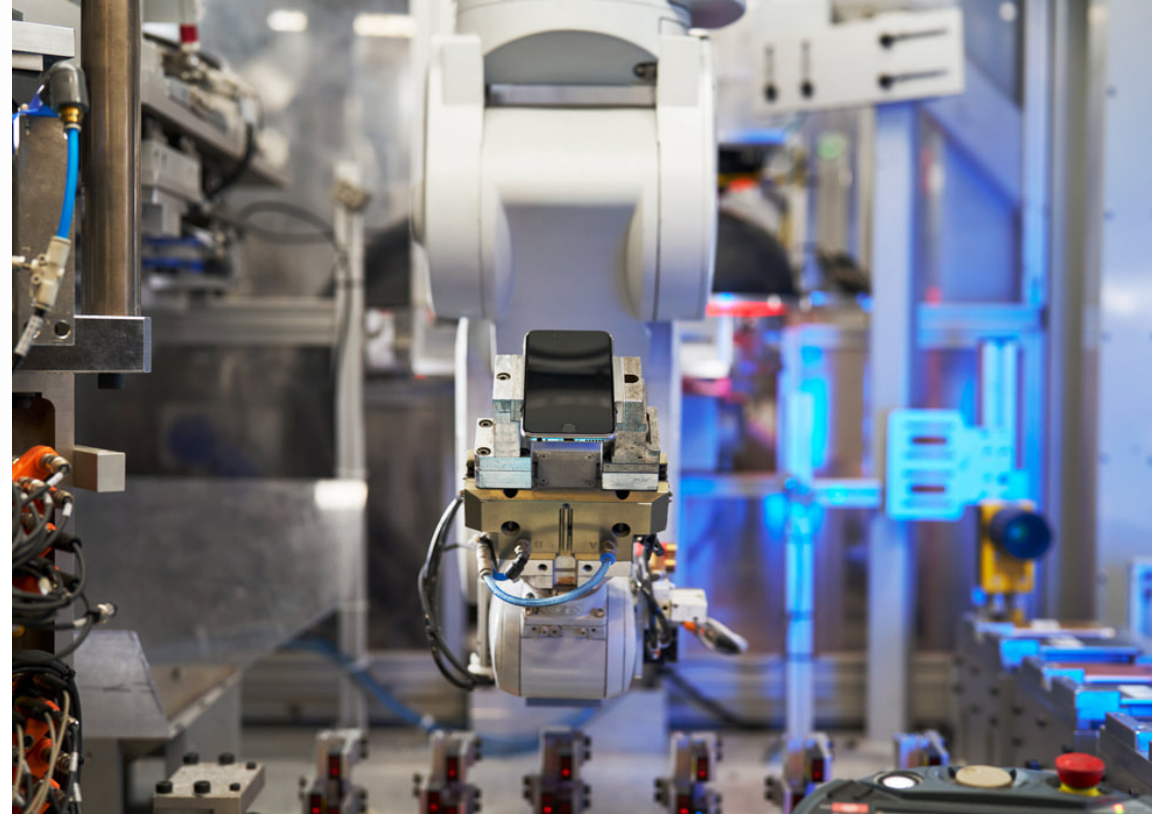
Why not just recycle everything at end of life?

How much of each material in electronics is [currently](#) recycled?

- Iron (used both in small-scale electronics and as a structural element in steel): 90%
- Aluminum (same as above): 85%
- Noble metals, silver, gold, and palladium: 80%
- Scarce metals: germanium, gallium, rare earth metals, tantalum: 1%
- Lithium, cobalt: 60-90%
- Plastics: unknown (!)

Recycling: metals

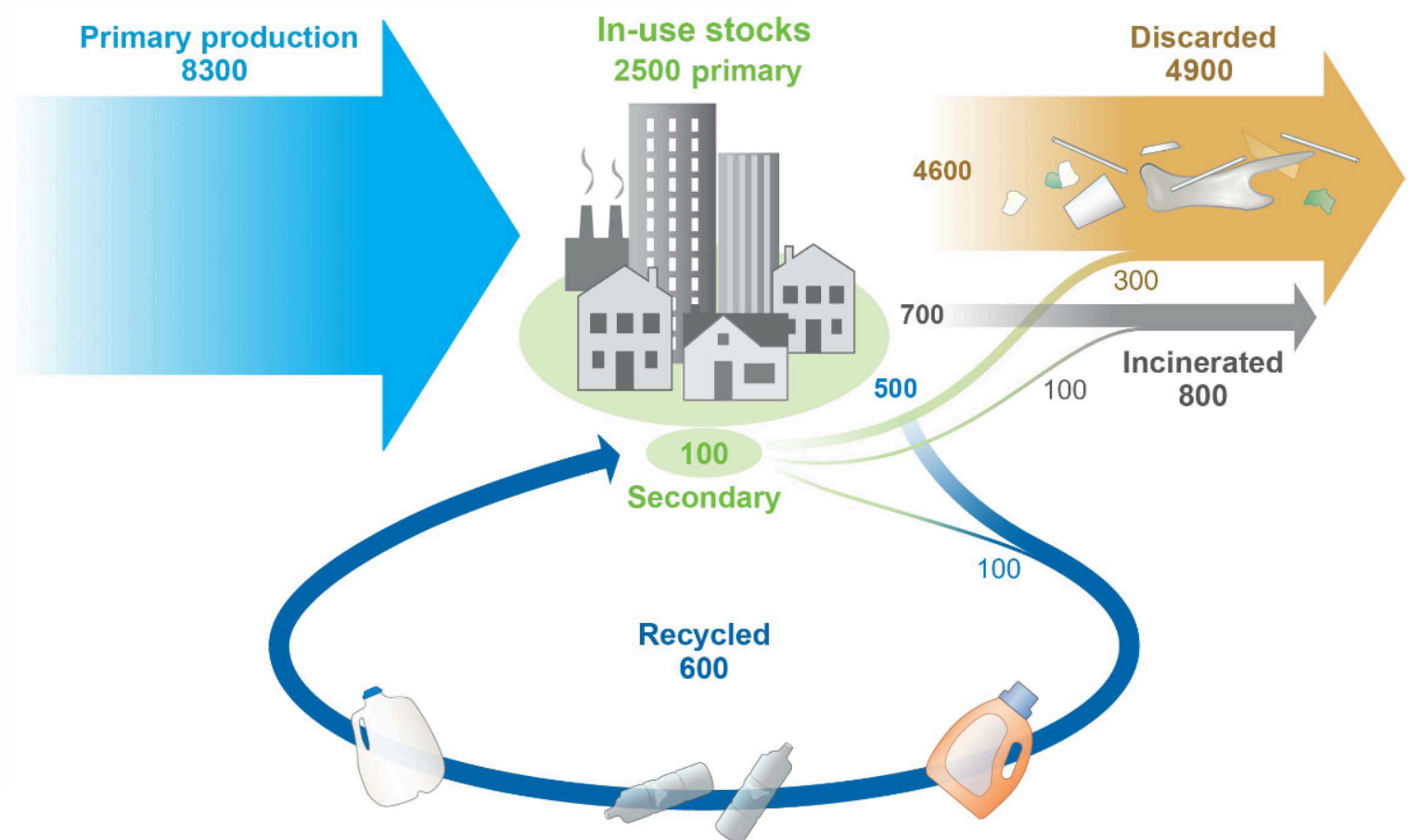
- Some metals have [higher recycling potential](#) than others; most aluminum worldwide is already recycled
- Tin, gold, silver frequently recycled due to economic value
- Rare earths [not recycled](#), due to increasing component miniaturization, thermodynamic limits of metallurgy



[Apple robot for device disassembly before recycling](#)

Recycling: plastics

- Almost completely produced from fossil fuels
- Vast majority cannot be recycled
- Obstacles: separating recyclable/unrecyclable plastics, high energy requirements, high costs, chemical degradation of feedstock
- Not a good idea



Global stocks and flows of plastics in human society
(megatons).

Recycling: batteries

- Technologies for 85-95% recycling rate [recently commercialized](#)
- Current end-of-life collection rates at [60%](#) and increasing fast, driven by sourcing concerns for lithium, cobalt, and nickel



A Changing chemistries

Switch to battery chemistries that use fewer or no critical battery minerals



B Higher energy density batteries

Pack more energy into a kilogram of battery, requiring fewer minerals to do the same work



C Recycling

Recover battery minerals at end of life to re-use in the manufacturing process and avoid the need for new minerals



D Reuse and extend lifetime

Use and reuse batteries for longer, to avoid having to make new batteries with new minerals



E Efficient vehicles

Improve vehicle efficiency and right-size cars for purpose to reduce the battery size per vehicle, and therefore mineral demand



F Efficient mobility

Improve urban planning, logistics efficiency, active modes, public transit, and electric micromobility to encourage alternate transport and lower EV mineral demand

[Strategies to ensure maximum mineral use and recovery](#)

Designing sustainable electronics

[Design principles](#) for sustainable electronics:

1. **Design for minimal material/energy footprint:** devices are [no more powerful/complex than task requires](#); fewer components → cheaper devices, rebound effects?
2. **Design for elimination of harmful substances:** find substitutes for toxic substances
3. **Design for repairability:** reduce demand for new devices by [enabling longer lifetimes](#); use standardized components; [avoid adhesives](#), non-replaceable batteries, proprietary screws
4. **Design for upgradability:** increase lifetime by allowing [component switching](#).
5. **Design for recycling:** use modular components, fasteners for [end-of-life disassembly](#).

Sustainable electronics: AirPods

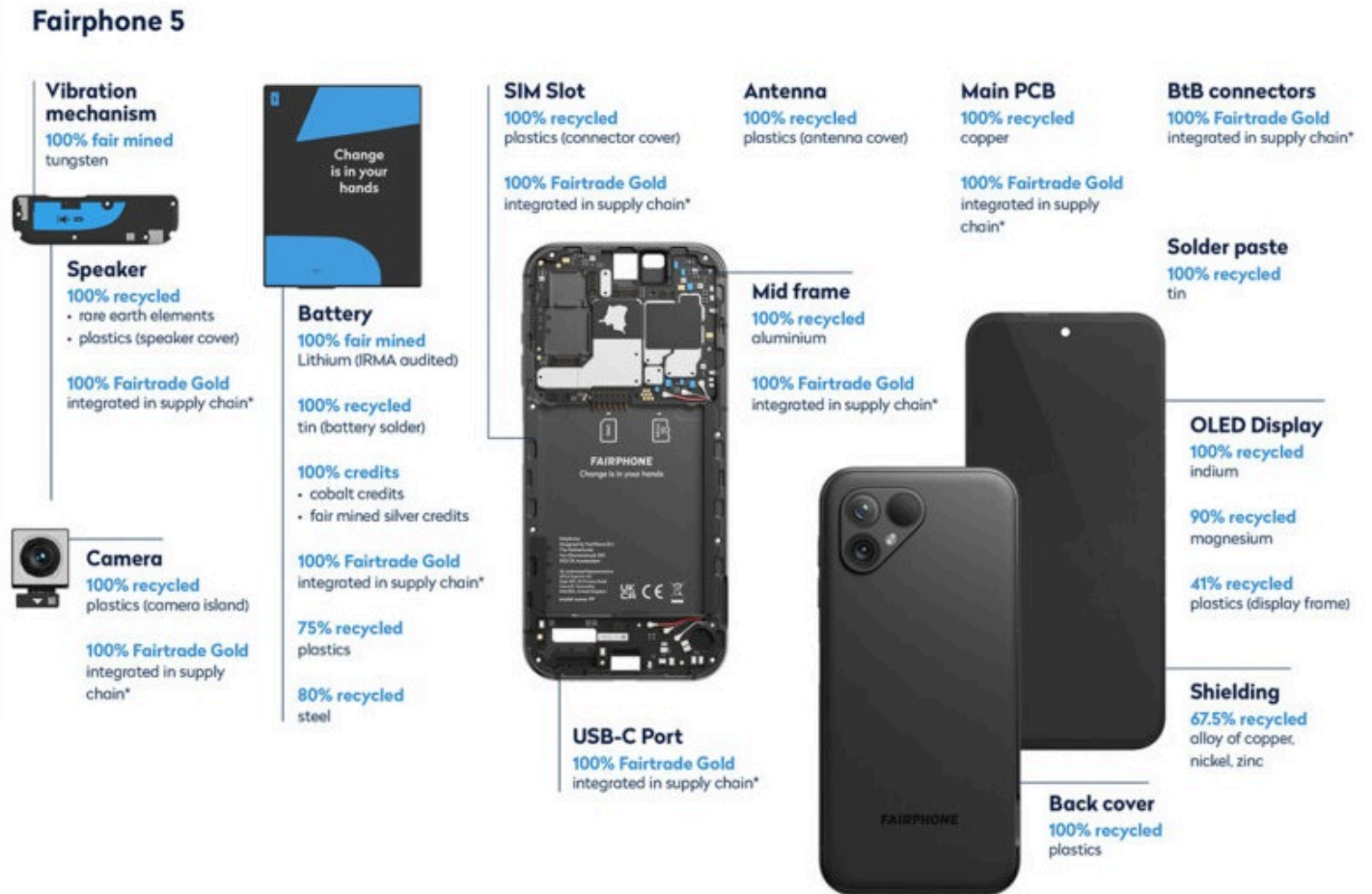
- True wireless Bluetooth [headphones](#):
 - Speakers: [transducers](#) (convert electrical signals into sound waves) [contain](#) diaphragm, copper voice coil, neodymium magnets
 - W1/H1 processor: purified silicon
 - Circuit boards: tantalum capacitors, gold pins
 - Casing: acrylonitrile butadiene styrene
 - Battery: rechargeable lithium-ion cell, [degrades](#) after [18-24 months](#)
- **Impossible** to repair without cutting apart outer casing: iFixit repairability score [0/10](#)



[We did it Patrick! We repaired the AirPods!](#)

Sustainable electronics: Fairphone

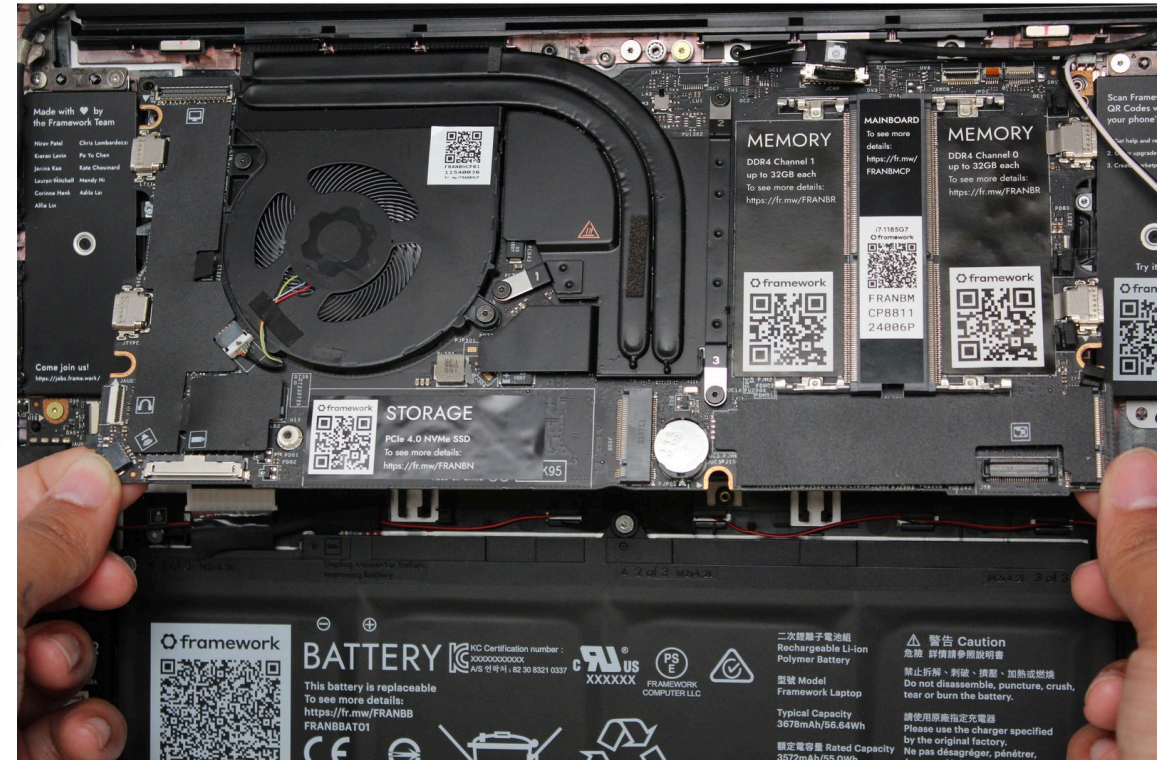
- Dutch electronics company aiming for sustainable products
- Modular device design: users can swap components on their own
- Maximum use of recycled materials, older phones repurposed for parallel computing research



'Fairly sourced', recycled materials in Fairphone supply chain

Sustainable electronics: Framework laptop

- US [startup](#) selling personal computers
- [Philosophy](#): well-considered design tradeoffs, trusting customers/repair shops with access/information → good, easy to repair devices
- [Modular device design](#): repairs/upgrades increase device lifetime
 - Captive fasteners
 - Backwards compatible hardware design
 - Replacement parts sold separately
 - Mainboard as [standalone module](#), can be used as single-board computer

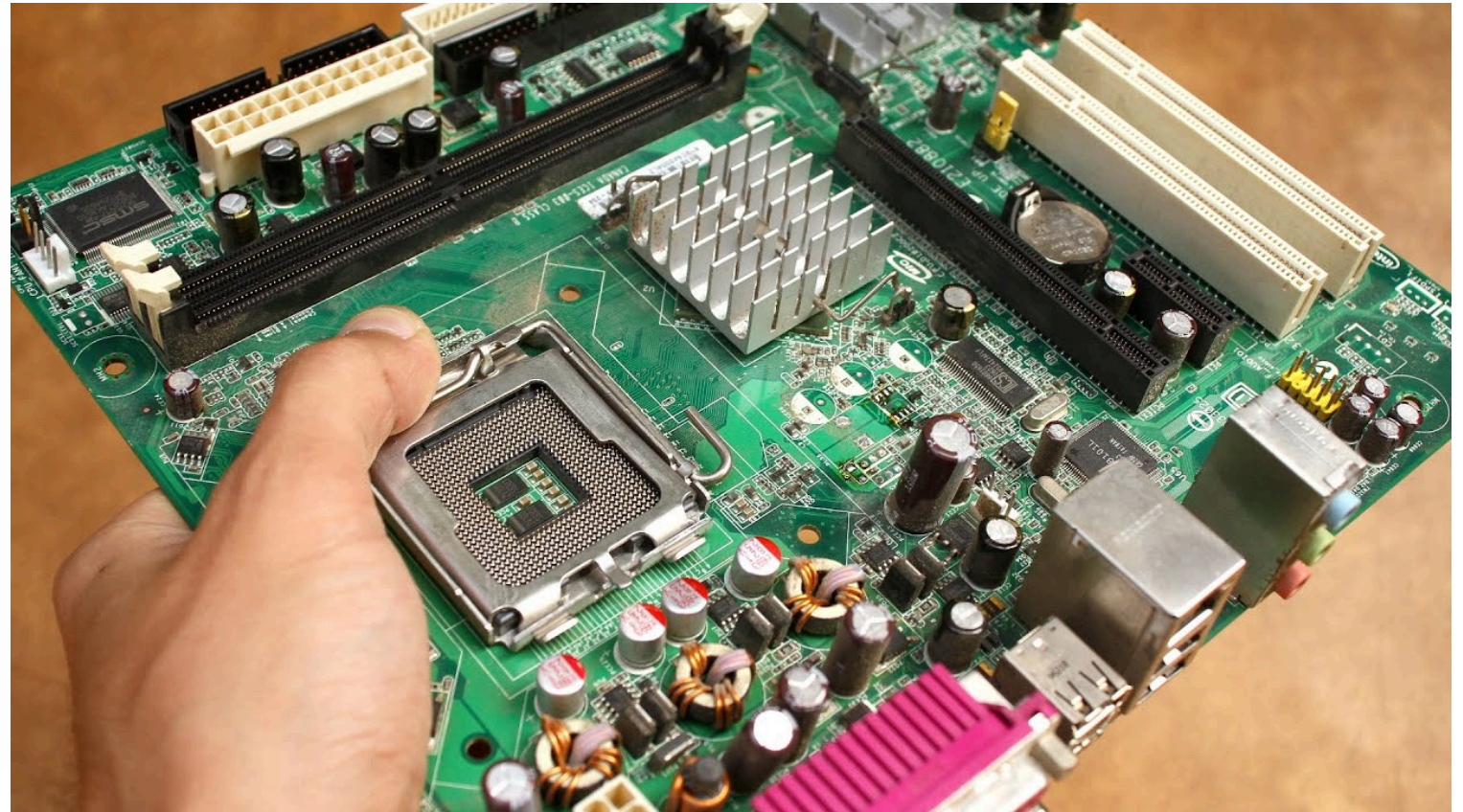


[Replacing the mainboard of a Framework laptop](#)

How do we design sustainable electronics in practice?

PCB design: history

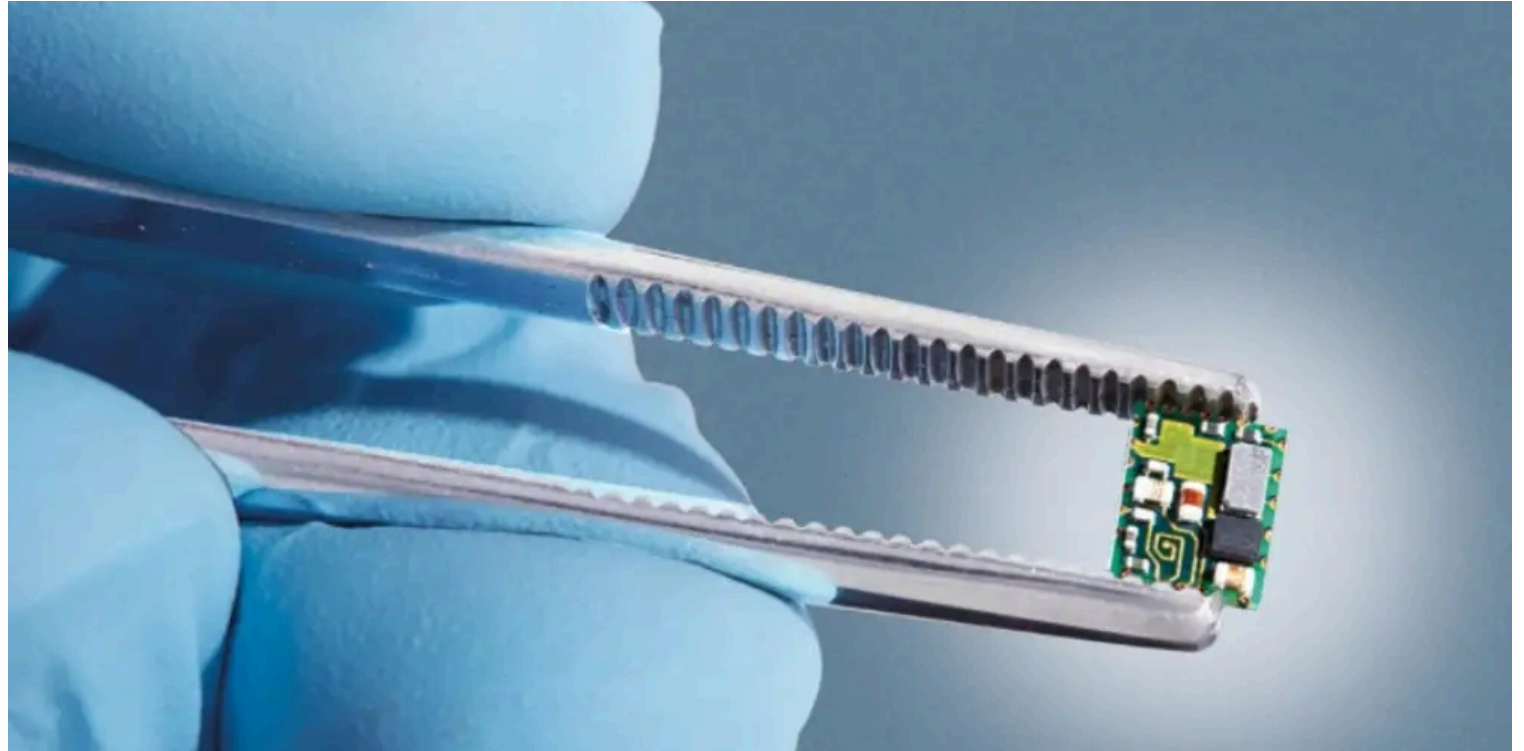
- Older motherboards feature both soldering, mechanical fasteners (e.g. [retention lever](#))
- Large components allow for manual depopulation (removal of components)
- Components reused or recycled as desired



[Mechanical connectors on older motherboards](#)

PCB design: current trends

- [Miniaturization](#): smaller components, smaller PCB's
- No longer possible to manually depopulate
- At end of life, PCB's [discarded](#) into E-waste stream

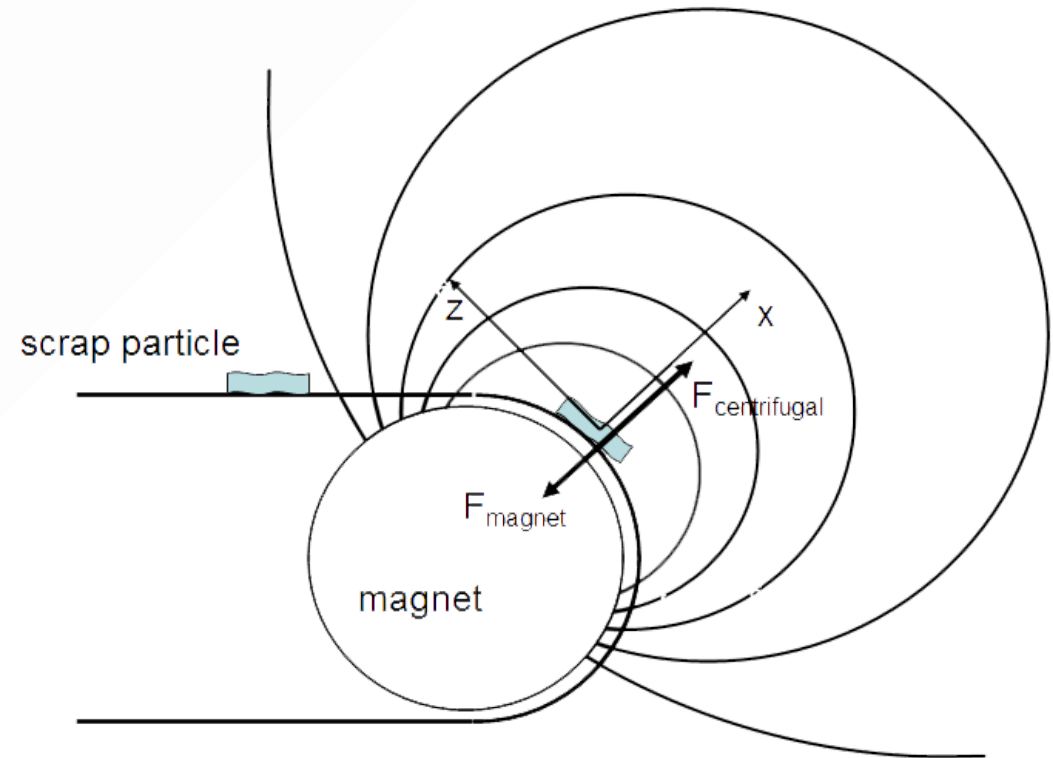


You may not believe it, but this PCB is perfectly average

PCB design: typical recycling methods

- [Visual sorting](#): computer vision algorithms to identify PCB components
- [Magnetic sorting](#): ferromagnetic sorting of PCB components in clean scrap machine
 - Pursued at [TU Delft](#)!
 - Still requires manual depopulation!
- [Mechanical recycling](#): shredding PCB's, separating metal powder from plastic
- [Chemical recycling](#): submerging PCB's to exfoliate e.g. gold plating

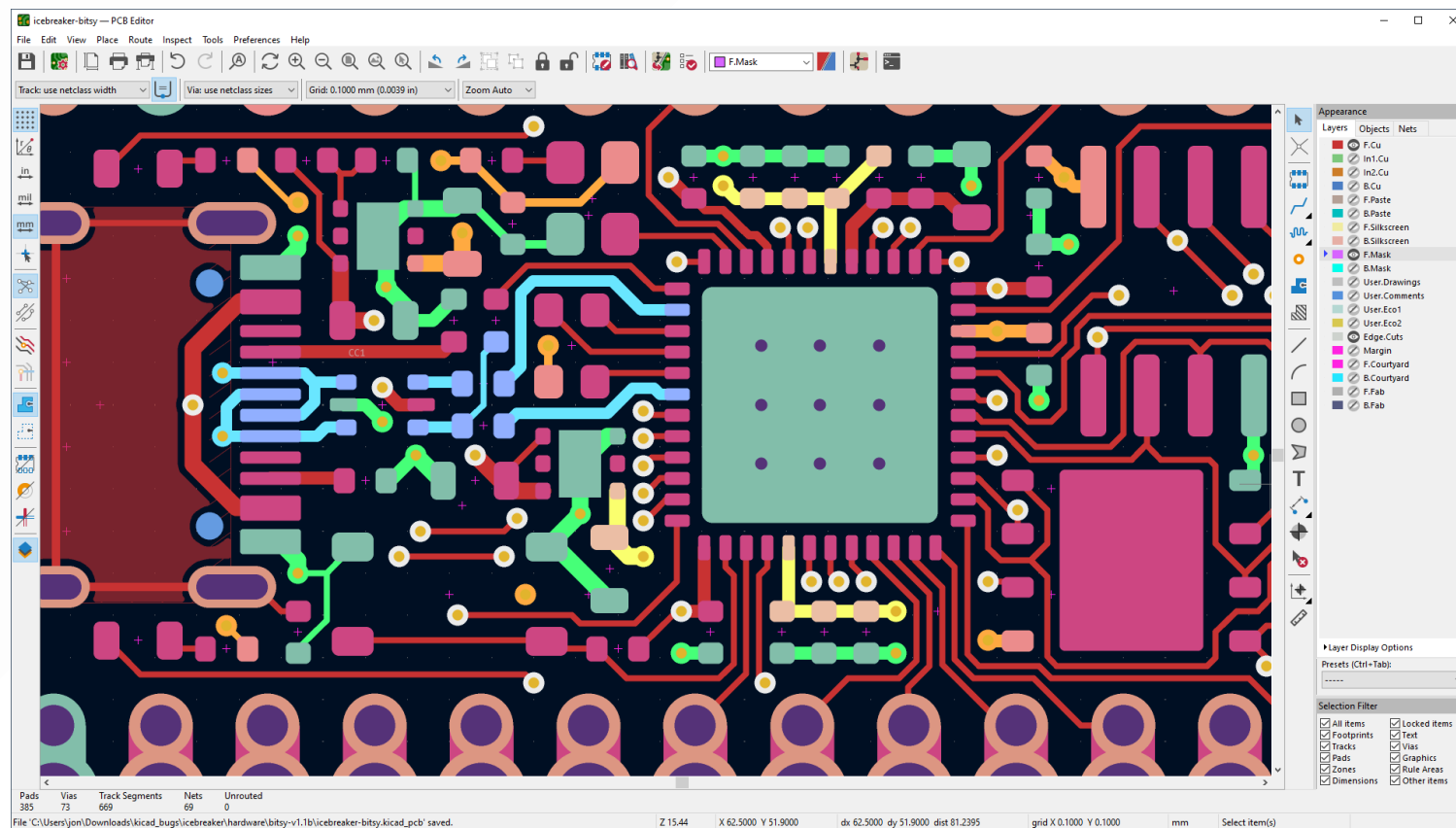
No method is sufficient on its own!



[Ferromagnetic sorting: components sorted by metal composition](#)

PCB design: design rules

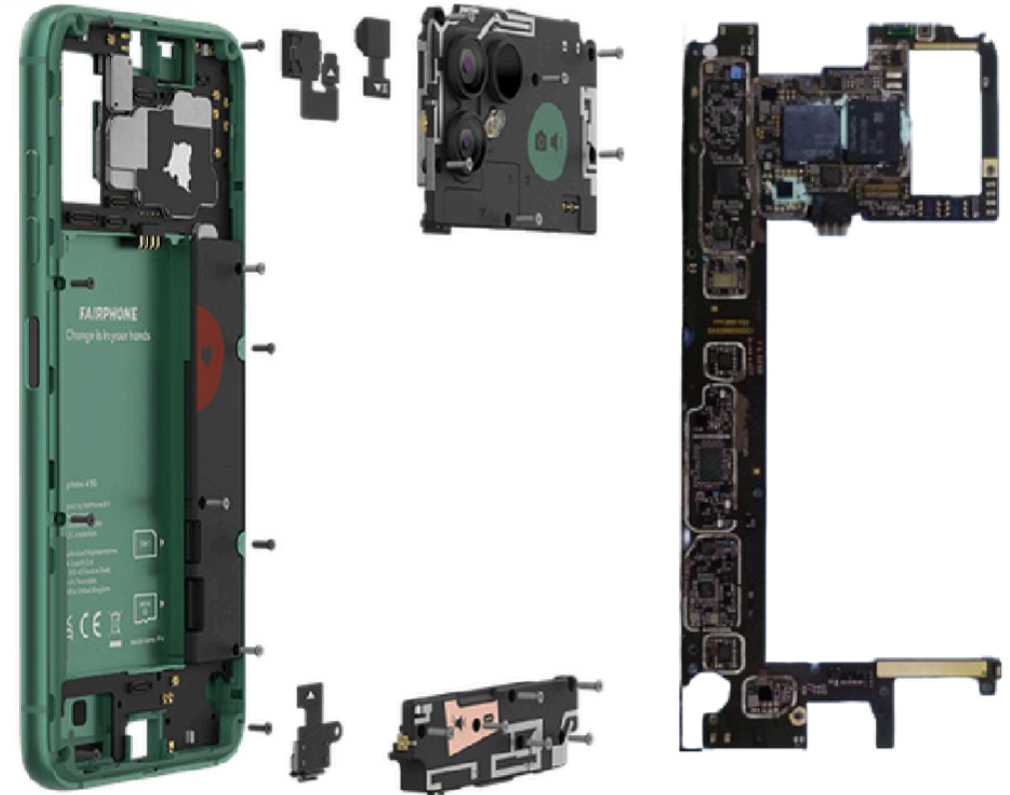
- Component reuse preferable to recycling → explicitly design PCB for eventual depopulation
- [Design rules](#): provided by manufacturers to ensure PCB can be produced
- [Custom design rules](#): set by designer in e.g. [KiCad](#)
 - What would a design rule to allow for manual depopulation look like?
 - What would a design rule for automatic PCB dissassembly look like?



[PCB layout design in KiCad](#)

PCB design: DeltaLCA

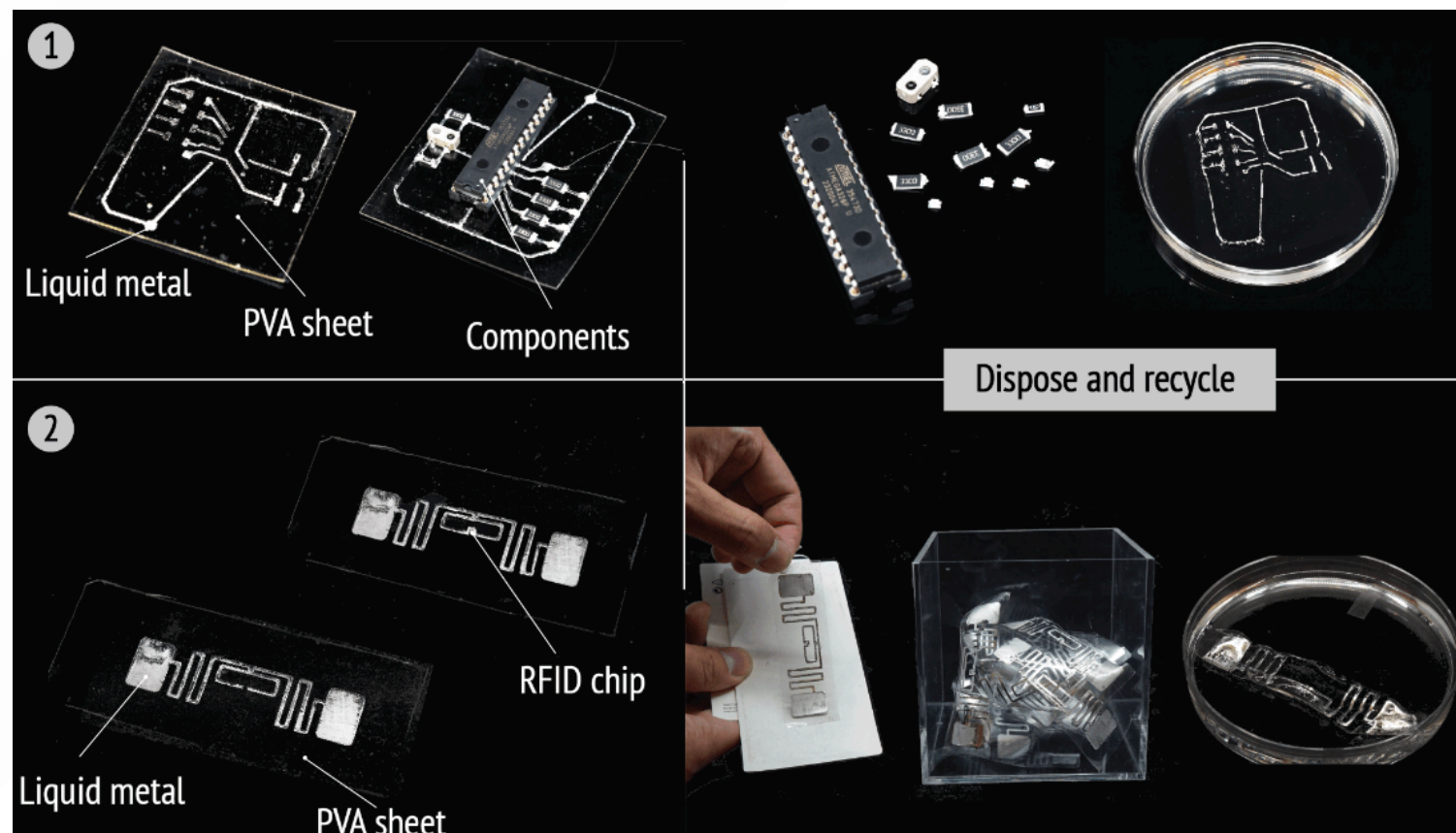
- [Difficult](#) to perform lifecycle analysis for electronics: only happens after manufacturing, hundreds of components, incomplete information, no clear recommendations for design
- [DeltaLCA](#): methodology for comparing environmental impact of PCB designs even with incomplete data:
 1. Compile database of standard components and known impacts
 2. Estimate known impacts of each design based on components
 3. Use domain-specific heuristics (e.g. larger diodes require more materials) to compare unknown impacts



[The Fairphone 4 has over 300 components](#)

PCB design: Recy-tronics

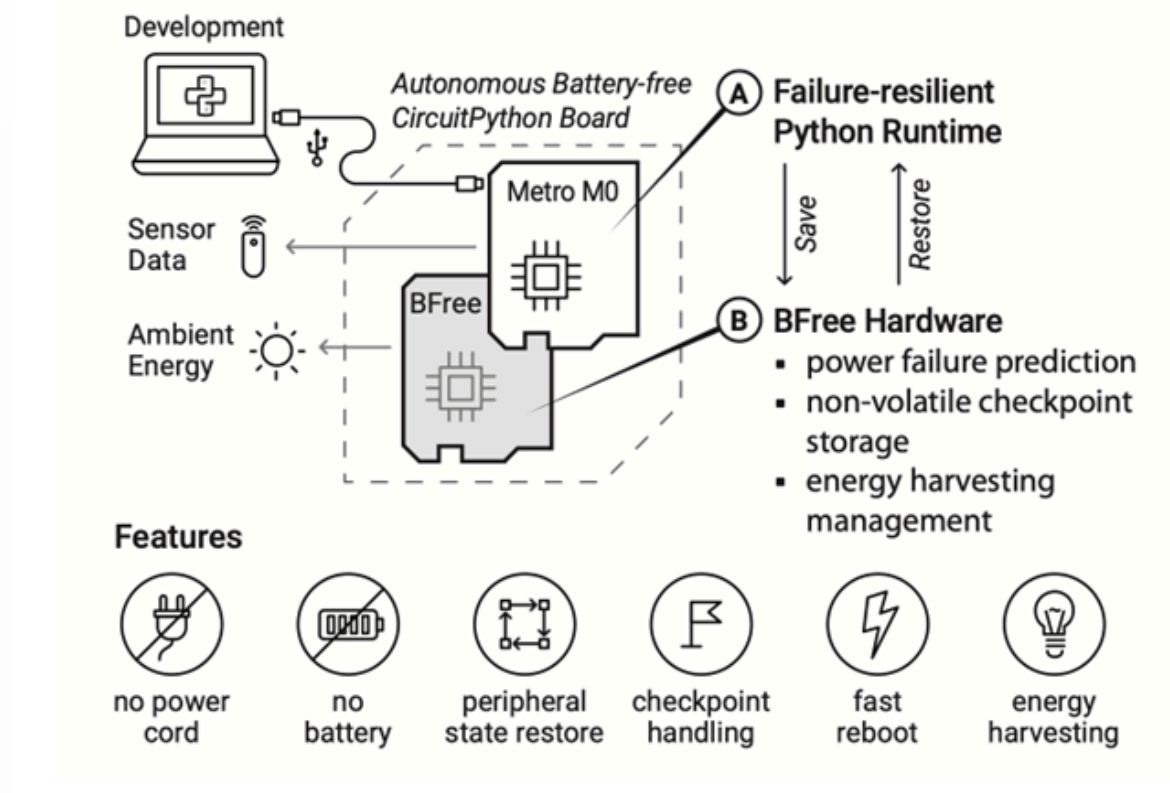
- [Design paradigm](#): use recyclable materials with versatile form factors
 - Polyvinyl alcohol (PVA) film: soluble in water, easy to recycle
 - Liquid metals (RTLMs) for flexible circuits
- [Example](#): proximity sensors, RFID clothing price tags
 - Recycling: submerge in water, PVA dissolves, recover components



[Recy-tronics can be dissolved in water to recover components](#)

Device design: battery-free prototyping

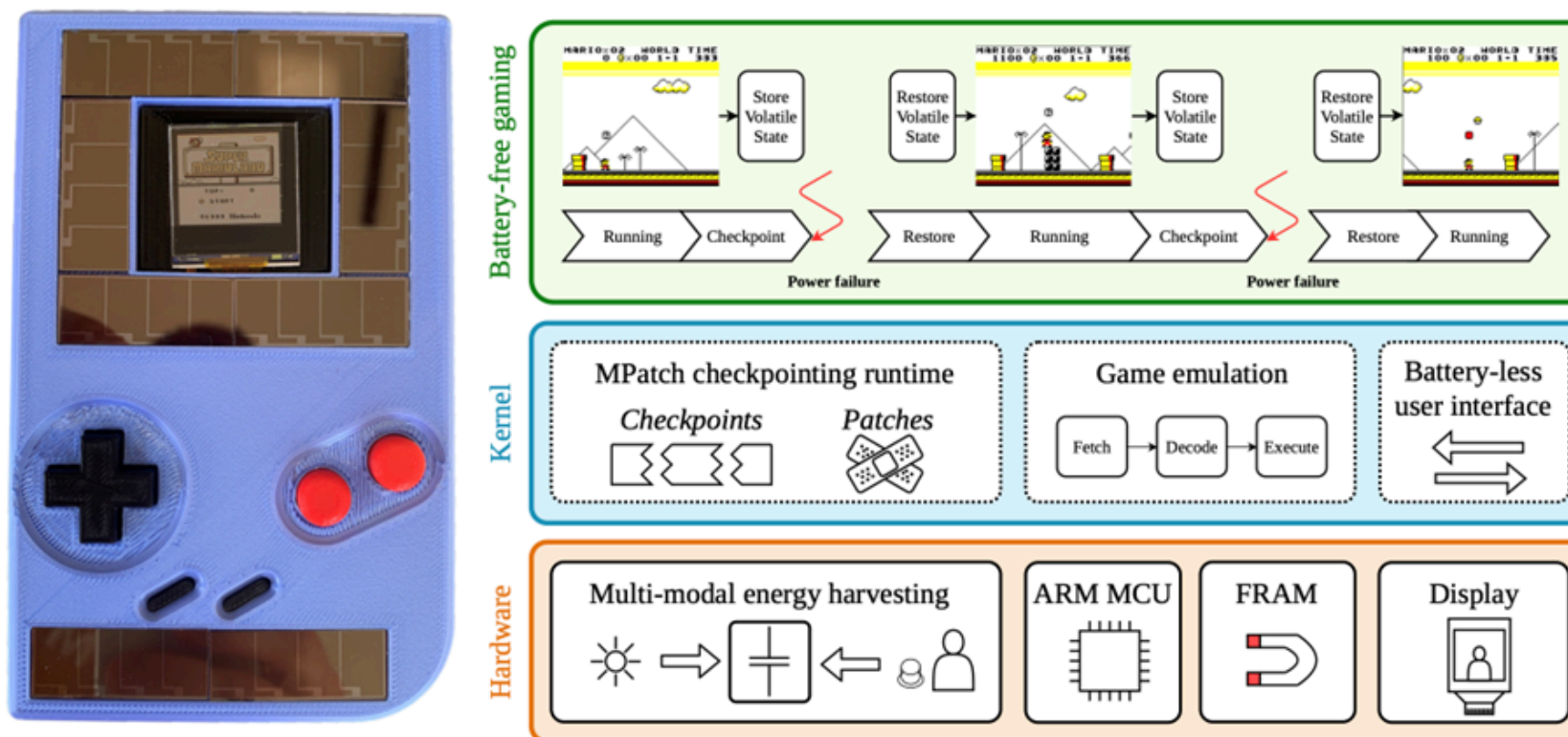
- [Design paradigm](#): removing batteries from Arduino-style prototyping platforms



[BFree system diagram](#)

Device design: battery-free Game Boy

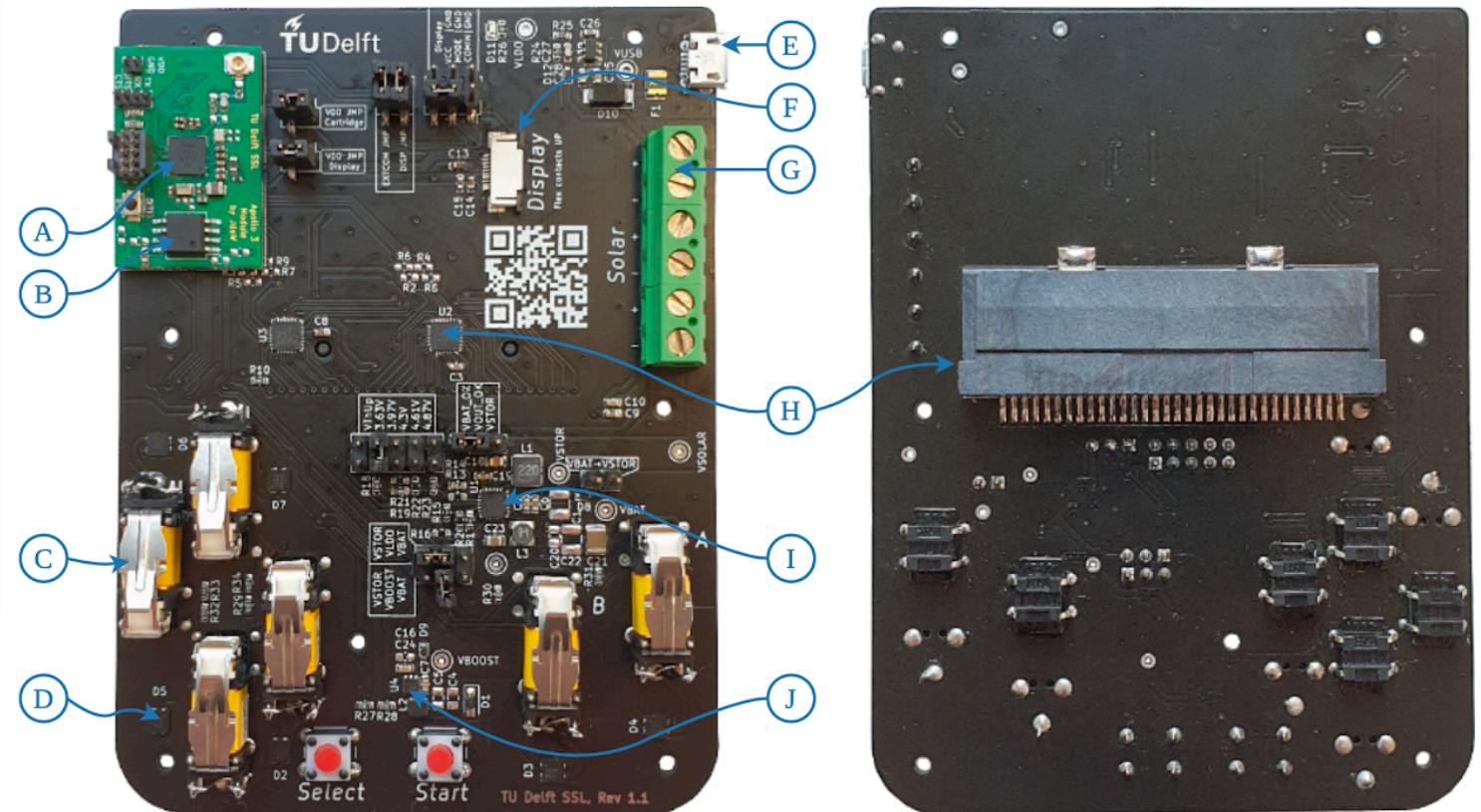
- [Design paradigm](#): removing batteries from handheld game platforms



[Battery-free Game Boy system architecture](#)

Device design: battery-free Game Boy

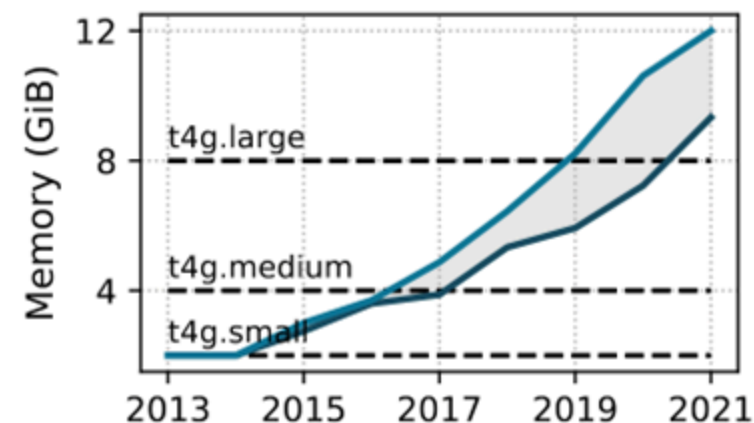
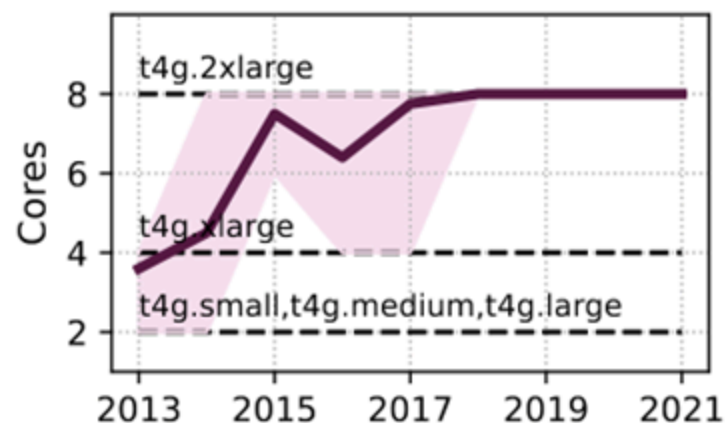
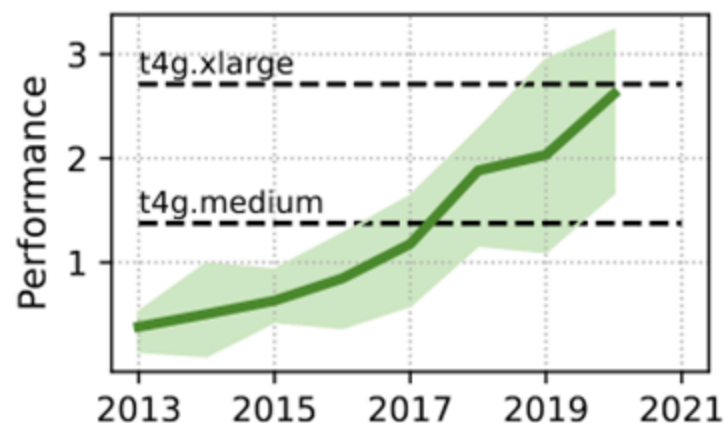
- Custom PCB design: low-power MCU, solar cells, energy generating buttons, capacitor for transient energy storage, low power screen
- What is the lifecycle impact compared to a regular Game Boy?



Battery-free Game Boy system PCB layout

Device design: junkyard computing

- Millions (billions?) of smartphones manufactured each year, old devices discarded or left to gather dust
- [Design paradigm](#): reuse of obsolete devices (Google Pixel 3A) for cloud computing
- Cloudlet of Pixel 3A's can successfully perform cloud microservices

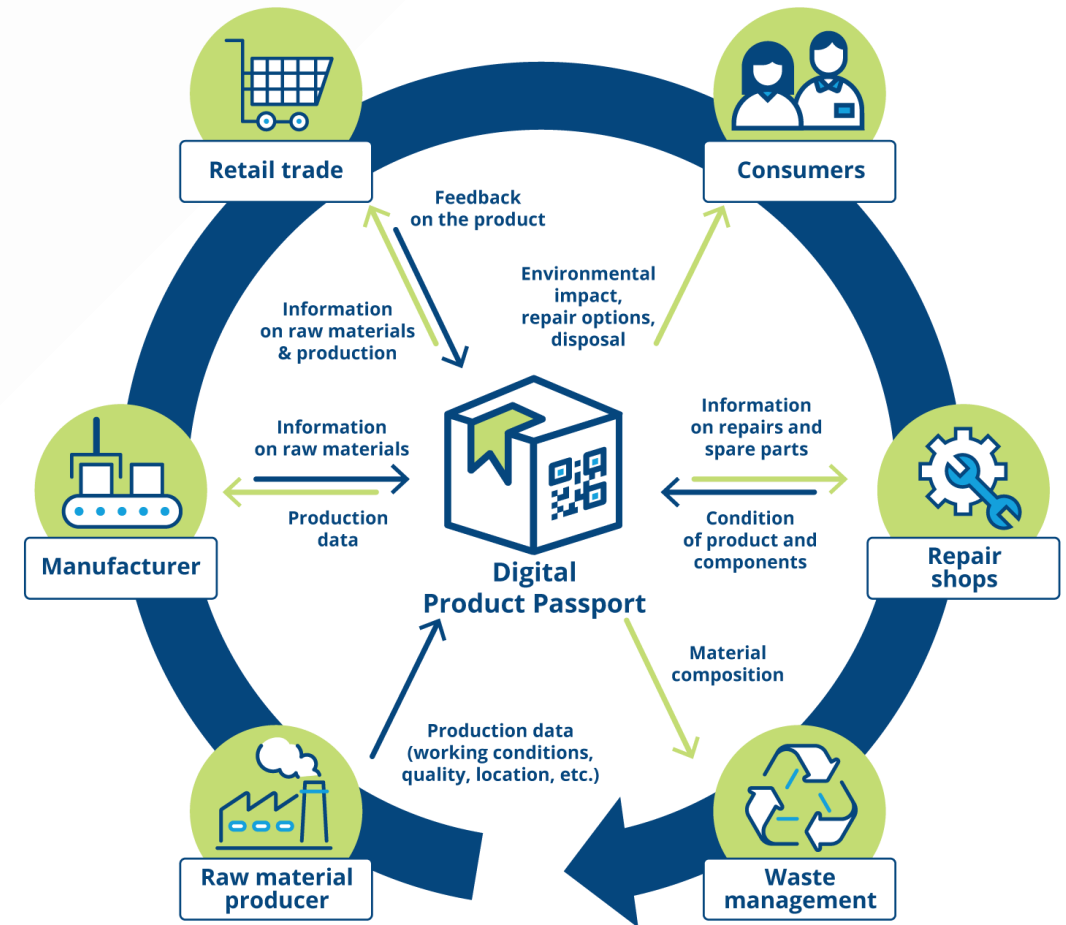


[Capabilities of smartphones compared to Amazon t4g cloud computing](#)

How do we manage electronics sustainably?

Digital product passports

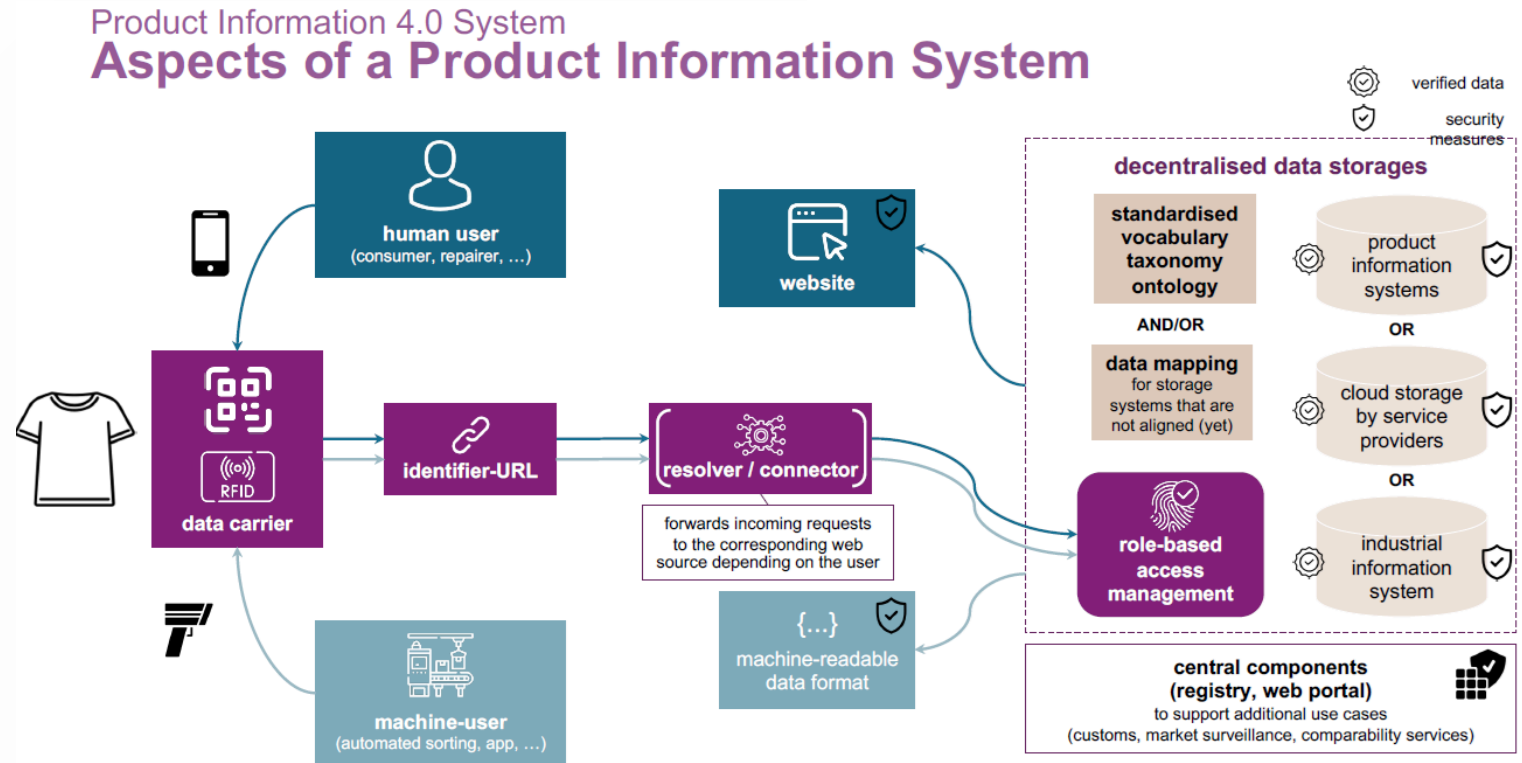
- [Digital product passport](#) (DPP): data on composition (materials), origin, and environmental impact of products (reusability, energy consumption, disposal)
- Various EU/UN regulatory frameworks, many competing data formats
- [In the EU](#): applied to textiles, batteries, electronics
- Objective: [provide FAIR](#) (findable, accessible, interoperable, reusable) data, understood both by humans and by machines



Typical information in a DPP

Digital product passports: components

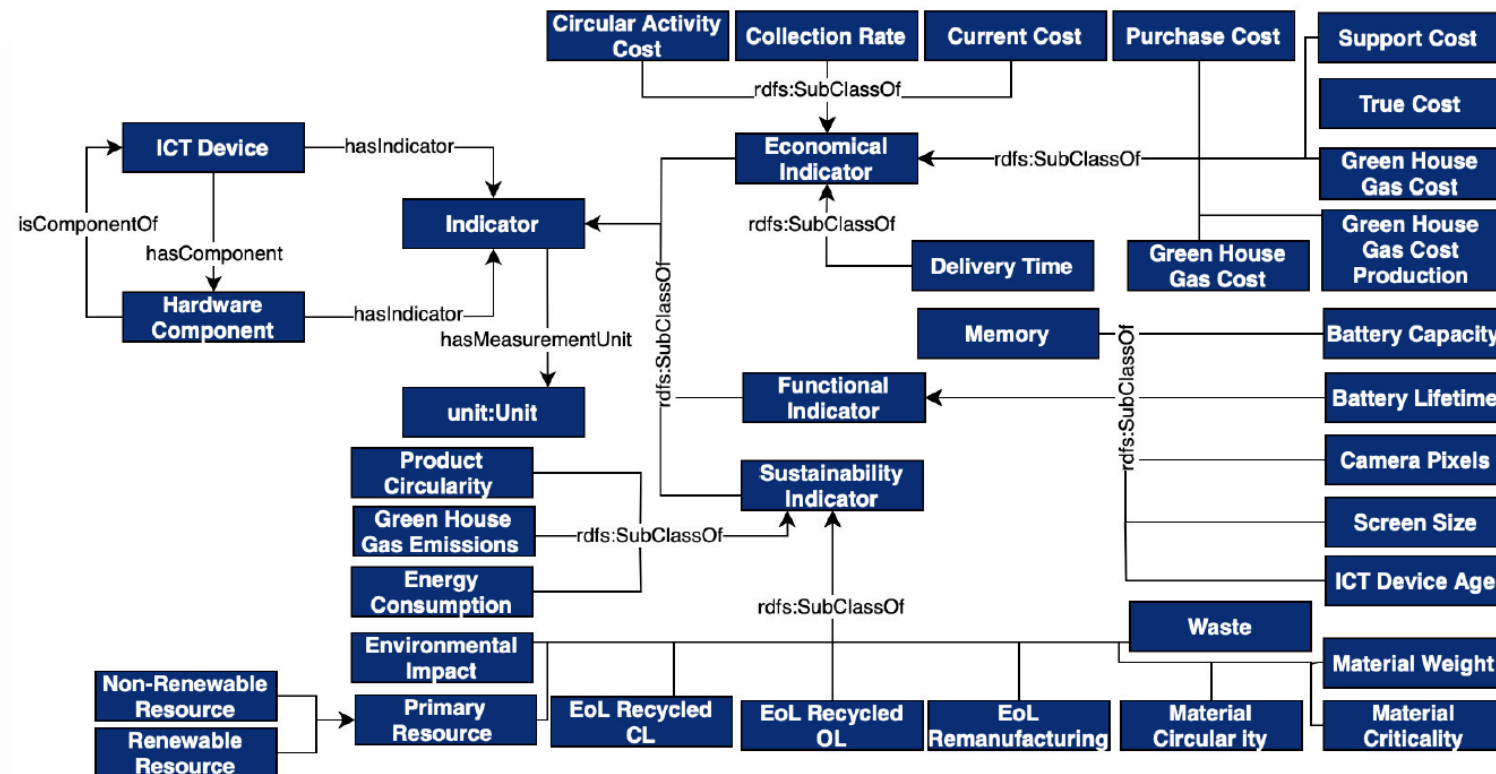
- DPP acts as a product information system:
 - Data carrier (e.g. RFID)
 - Unique identifier-URL
 - Resolver
 - Data storage
- Data adds up throughout product lifecycle:
 1. Producer creates DPP
 2. Use and repair data added to DPP
 3. At end-of-life, DPP is destroyed



Digital product passport as a product information system

Digital product passports: RePlanIT

- TU Delft implementation of DPP for digital devices as [ontology and knowledge graph](#)
- Representing device properties (e.g. repair schedule, components, materials) as classes, subclasses, and indicators



[RePlanIT representation of a laptop](#)

Digital product passports: criticism

- Most DPP applications still in research or pilot phase (e.g. [CIRPASS](#) project)
- Large expected data volumes → centralized or decentralized data storage?
- Underlying assumption: better information helps companies make better decisions about reducing emissions, waste → is this really true?

Recap

- Electronics contain a broad range of materials and elements, we cannot rely on recycling
- Explicit design for minimal footprint, repairability, upgradeability, recyclability required
- PCB's currently designed for computing density, need to shift to design for depopulation
- Creativity required in how we design and reuse hardware: eliminating batteries, reusing obsolete devices
- Digital product passports have interesting applications, but are in practice limited to research and pilots

Exam(ple) questions

1. **Why do AirPods not satisfy the principles of sustainable electronics design?**
 - a. They contain gold, which cannot be recycled
 - b. They contain a lithium-ion battery, which cannot be recycled
 - c. They are impossible to repair
 - d. They contain acrylonitrile butadiene styrene, which is toxic on contact
2. **What is the strongest obstacle to making electronics sustainable through recycling?**
 - a. The end-of-life collection rate for old batteries is too low to sustain mineral recycling
 - b. The vast majority of plastics in electronics cannot be recycled
 - c. Tin, gold, and silver components are too small to be effectively recycled
 - d. Battery minerals like lithium and cobalt are too cheap for recycling to be economical

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Feedback

- We need your feedback to continue improving the course!
- Feedback on Lecture 6 slides/notes:



Thank you for your attention!

Next lecture: Sustainable Data Storage, Part 1

Thursday, October 15, 15:45-17:30, EEMCS Hall F

- Drivers of data storage growth: consumer data, data retention, AI
- Data storage technologies: magnetic tape, optical discs, HDDs, SSDs
- Data storage architectures

Questions?